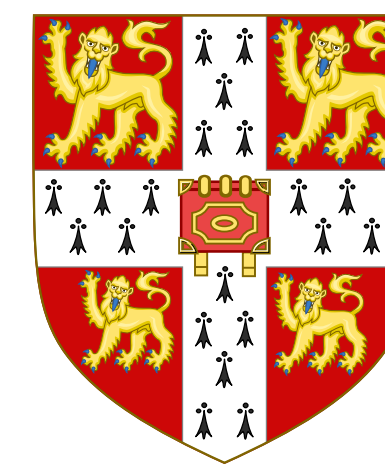




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What Is That Talk About? A **Video-to-Text** **Summarization Dataset** for **Scientific Presentations**

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TL;DR

We present **VISTA**, the first **scientific video-to-text** summarization dataset, and show that **plan-based method** improves quality and factual accuracy over strong multimodal baselines

Motivation

- Why is scientific video-to-text summarization important?
- Why do existing large multimodal models (LMMs) struggle with scientific videos?
- What are the limitations of current summarization approaches?
- How does this paper address these gaps?

Motivation

- *Why is scientific video-to-text summarization important?*
- Readers often prefer concise textual summaries to navigate dense video content
- Unlike entertainment or news videos, scientific content demands factual precision and structured reasoning

Motivation

- *Why do existing LMMs struggle with scientific videos?*
 - Most LMMs are tuned for general-domain videos (YouTube, movies) — not technical talks
 - No large-scale, domain-specific benchmark has supported evaluation and adaptation of models in this setting

Motivation

- *What are the limitations of current summarization approaches?*
- SOTA LMMs show problems with structural grounding → incoherence, hallucination

Motivation

- *How does this paper address these gaps?*
- VISTA (**V**ideo to **S**cientific **A**bstract) dataset
- Planning method

What is VISTA?

- 18,599 AI conference presentation videos paired with corresponding paper abstracts
- Covers top-tier venues (ACL, NeurIPS, ICLR, etc.)

Video

1

Factual knowledge

LM parameters encode a wealth of factual information

text-davinci-002

How many species of beetles are there?

There are more than 350,000 species of beetles.

350,000

Beetles (Order Coleoptera) are known to include some 350,000 described species. In the United States, there are nearly 30,000 kinds of beetles known. These figures are rising constantly due to the naming of new species by taxonomists. Size.

Smithsonian Institution

Beetles (Coleoptera) | Smithsonian Institution

2

PopQA

Focus: Factual knowledge

(Kathy Saltzman, occupation, Politician)

Subject Relationship Object

Task: Open-domain QA

Q: What is the **occupation** of Kathy Saltzman?

A: **politician**

3

Scale

Even the largest LMs barely outperform the smallest LMs for tail questions

Accuracy

Model parameters

1.3B 2.7B 6B 20B ~175B

GPT-neo OPT GPT-3 002 GPT-3 003

slope ≈ 0

Popularity

4

Complementarity of parametric and non-parametric memories

Retrieval is...

- especially helpful in the tail
- often harmful for popular knowledge

Not memorized in parameters → use retrieval

Memorized in parameters → don't use retrieval

What is Kathy Saltzman's occupation?

What is the capital of Louisiana?

unassisted LM retrieval-augmented

Popularity

Summary

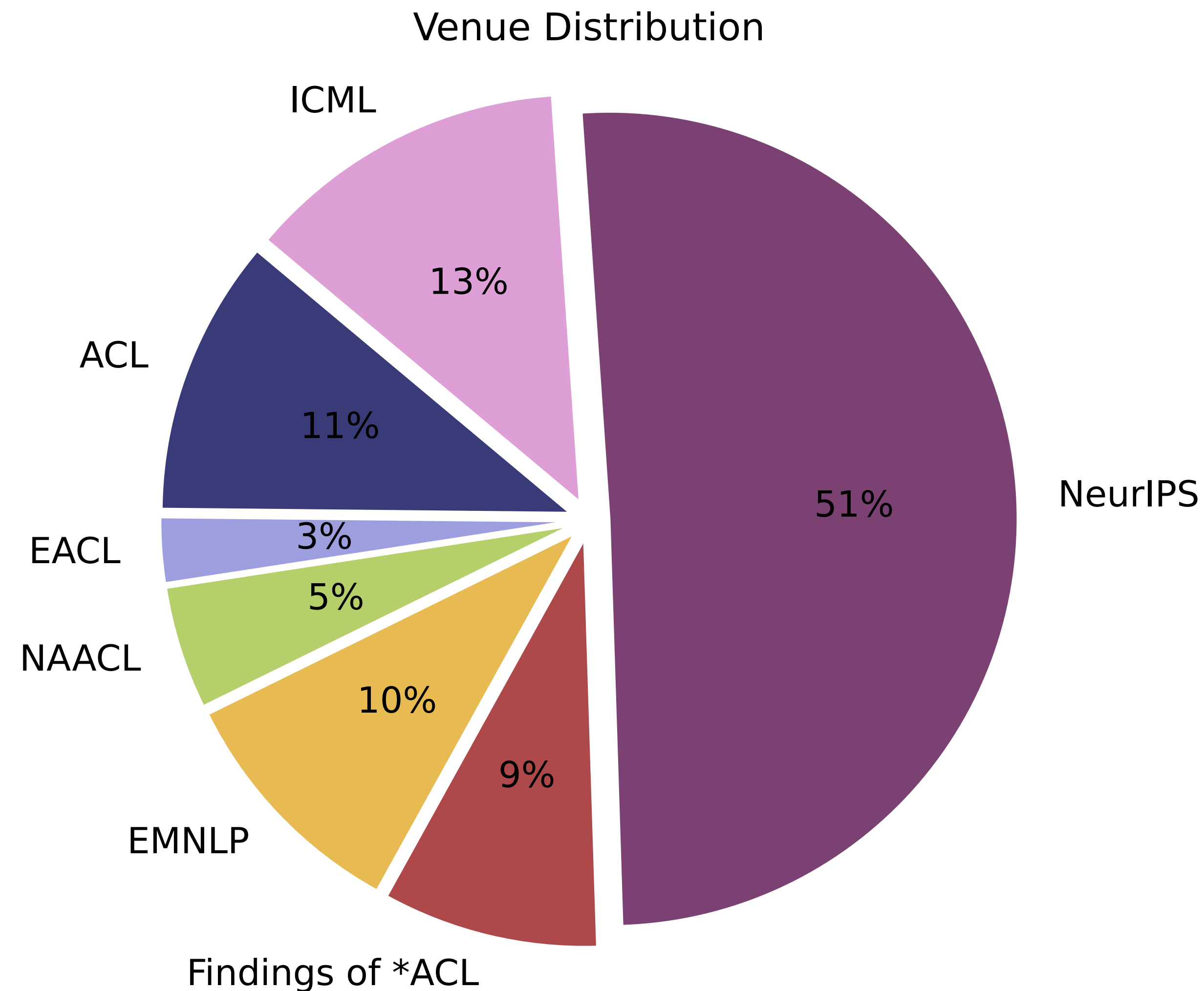
Despite their impressive performance on diverse tasks, large language models (LMs) [...], implying the difficulty of encoding a wealth of world knowledge in their parameters. This paper aims to understand LMs' strengths and [...], by [...]. We find that LMs struggle with less popular factual knowledge, and [...]. Scaling, on the other hand, mainly improves memorization of popular knowledge, and fails [...]. Based on those findings, we devise a new method for retrieval augmentation[...] memories when necessary.

Quality Control

- Manual check: 500 random pairs reviewed by two PhDs — 0 rejections
- Automated check: GPT-o1 flagged 39 samples, all confirmed valid by manual review
- Quality criteria: *Each summary must be concise and accurately reflect the video content*

Dataset Split

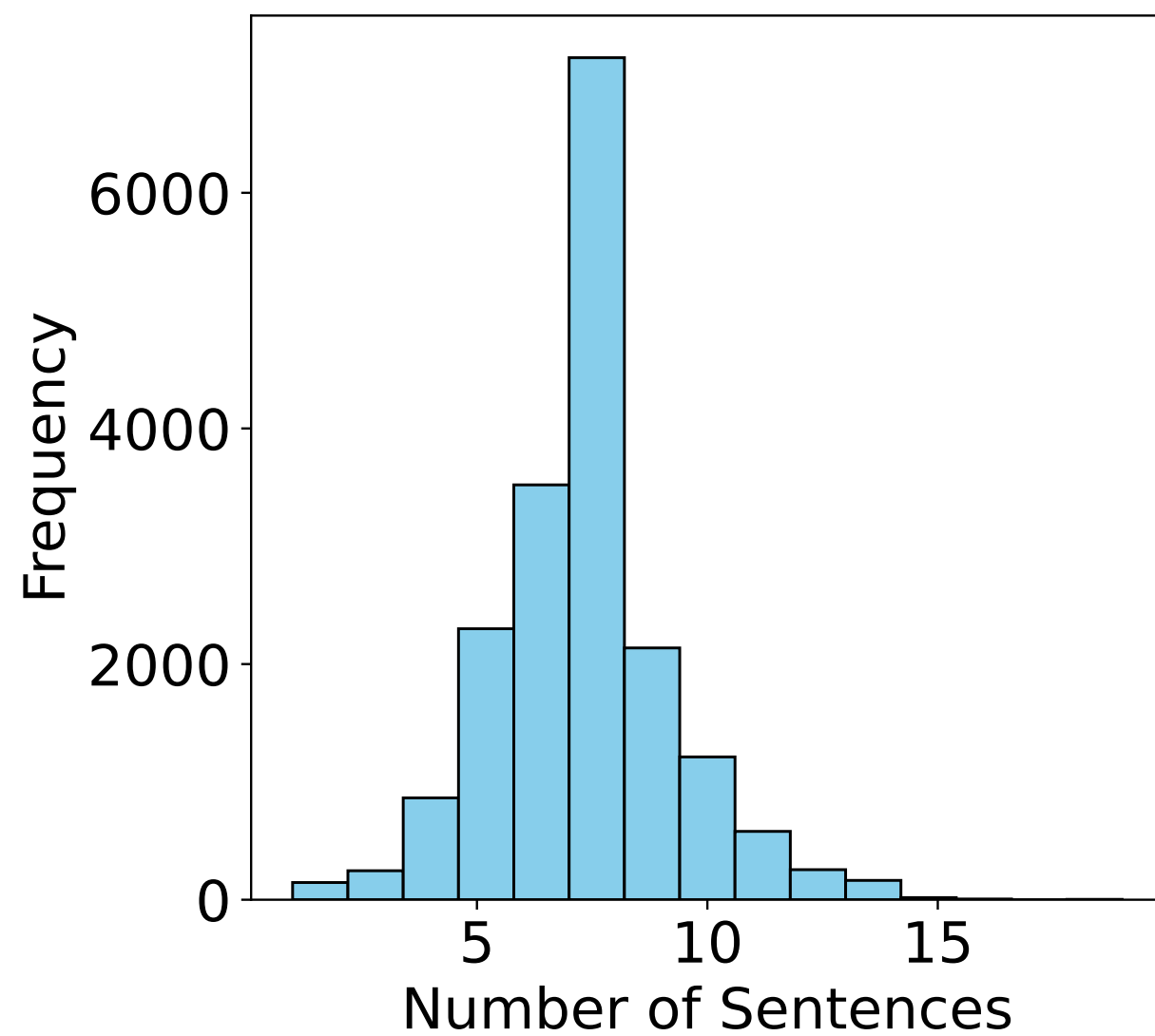
- Data Split:
 - Training 80%, Validation 10%, Test 10%
- Filtering:
 - Only paper presentations (no tutorials/invited talks)
 - Videos: 1–30 min, English, 1-to-1 paper alignment



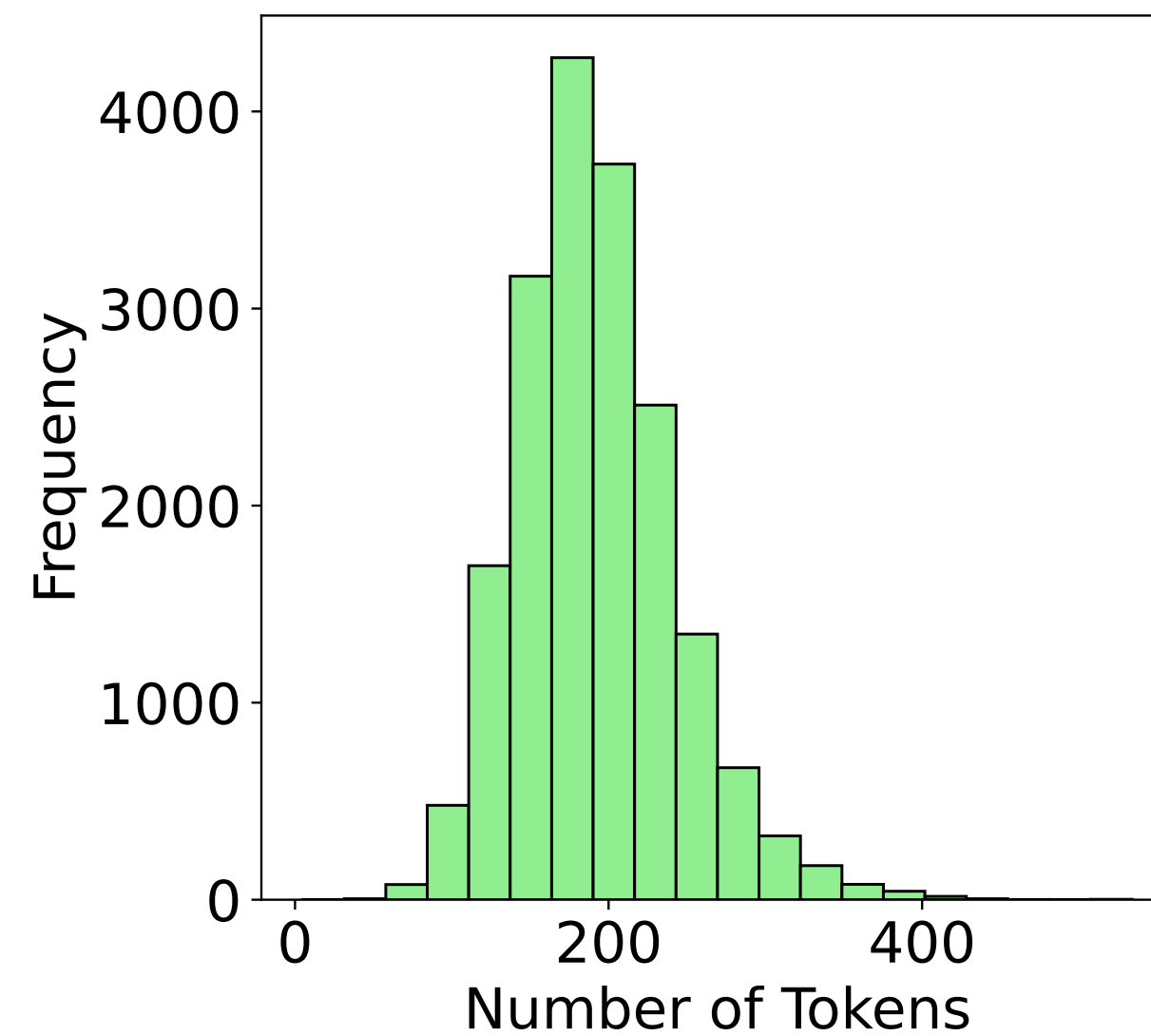
Dataset Statistics

- Most summaries remain under 250 tokens and 10 sentences
- Most videos last fewer than 10 minutes with under 30 shots

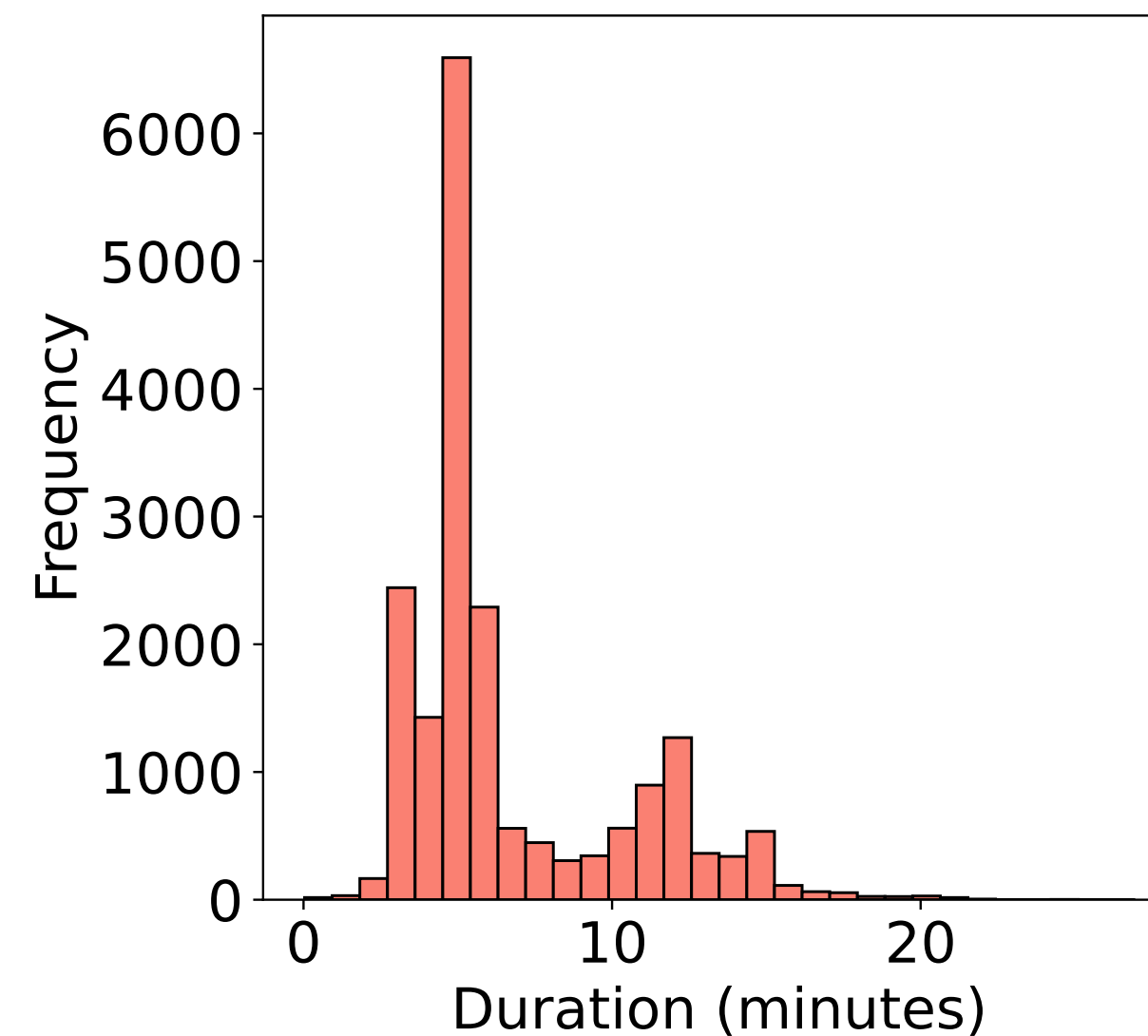
Sentence Count Distribution



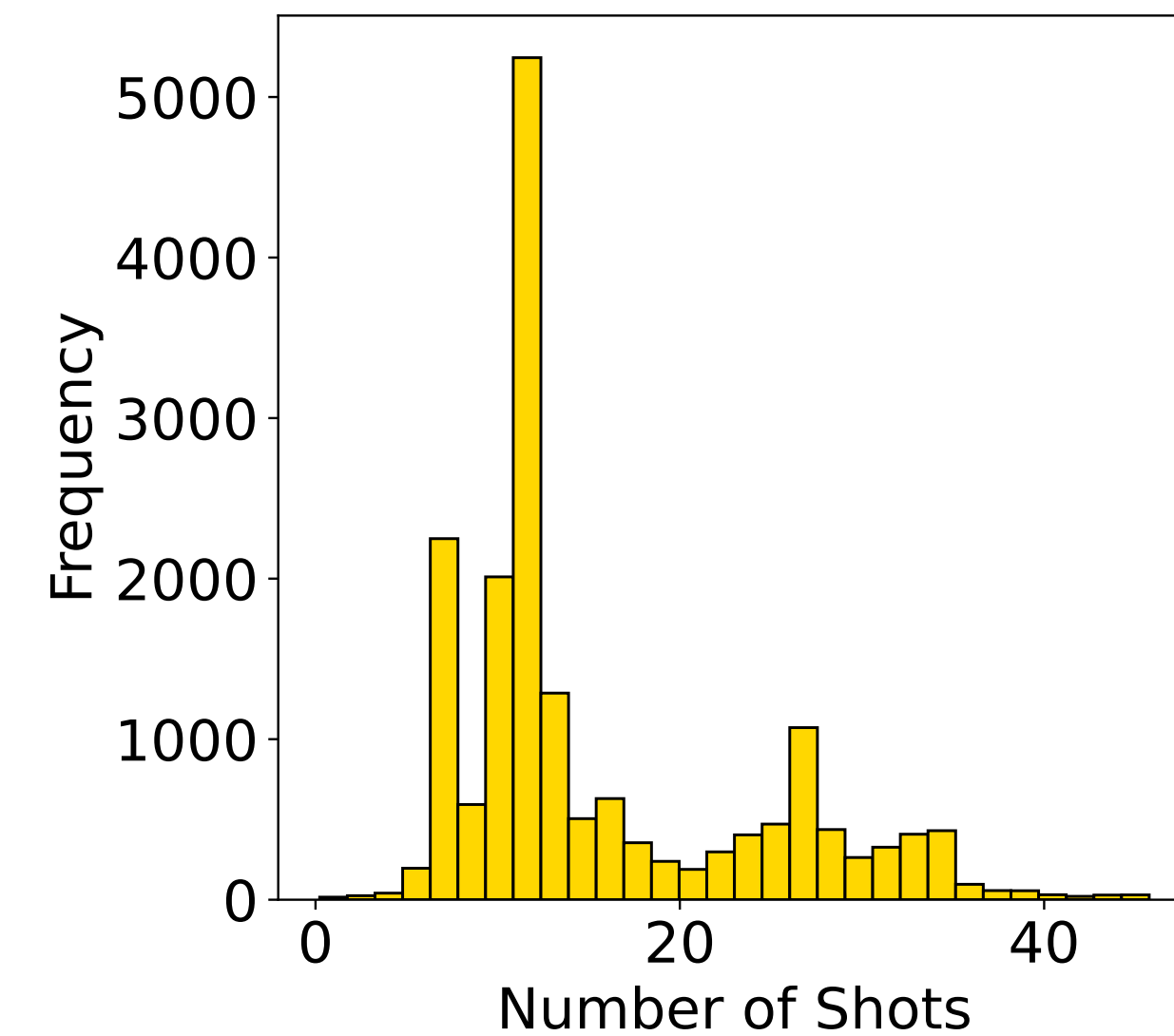
Token Count Distribution



Video Duration Distribution



Video Shots Distribution



Dataset Comparison

- **✗** Existing datasets: short clips, casual topics (e.g., VideoXum, YouCook2, etc.)
- **✗** Prior datasets focus on narrations, actions, or subtitles

Dataset	Language	Domain	#Videos	VideoLen	SumLen
MSS (Li et al., 2017)	English, Chinese	News	50	3.4	—
YouCook2 (Zhou et al., 2018)	English	Cooking	2.0K	5.3	67.8
VideoStorytelling (Li et al., 2019)	English	Open	105	12.6	162.6
VMSMO (Li et al., 2020)	Chinese	Social Media	184.9K	1.0	11.2
MM-AVS (Fu et al., 2021)	English	News	2.2K	1.8	56.8
MLASK (Krubiński and Pecina, 2023)	Czech	News	41.2K	1.4	33.4
VideoXum (Lin et al., 2023)	English	Activities	14.0K	2.1	49.9
Shot2Story20K (Han et al., 2025)	English	Open	20.0K	0.3	201.8
BLiSS (He et al., 2023)	English	Livestream	13.3K	5.0	49.0
SummScreen ^{3D} (Papalampidi and Lapata, 2023)	English	Open	4.5K	40.0	290.0
Ego4D-HCap (Islam et al., 2024)	English	Open	8.3K	28.5	25.6
Instruct-V2Xum (Hua et al., 2024)	English	Open	30.0K	3.1	239.0
MMSum (Qiu et al., 2024)	English	Open	5.1K	14.5	21.7
LfVS-T (Argaw et al., 2024)	English	YouTube	1.2K	12.2	—
VISTA (ours)	English	Academic	18.6K	6.8	192.6

Benchmarking

- **Closed-source LMMs:** GPT-o1, Gemini 2.0, Claude 3.5 Sonnet
- **Open-source video LMMs:** Video-LLaMA, Video-ChatGPT, Video-LLaVA, LLaVA-NeXT, mPLUG-Owl3
- **Text baseline:** LLaMA-3.1 (transcript, OCR)
- **Audio baseline:** Qwen2-Audio

Experiment Settings

- Learning Settings:
 - Zero-shot inference, QLoRA fine-tuning, Full-parameter fine-tuning
- Training Details:
 - Standardized hyperparameters (AdamW, learning rate = $5e-5$, batch size = 16, 16 epochs, early stopping)
- Video Preprocessing:
 - Video frames sampled at 0.1 fps, 32 frames per video
 - Transcription via Whisper, OCR via EasyOCR for text baselines

Evaluation Metrics

- Automated Metrics
 - ROUGE, SacreBLEU, METEOR, BERTScore, CIDEr-D
 - VideoScore: text–video alignment
 - FactVC: factual consistency
- Human Evaluation
 - 50 randomly sampled test videos
 - 3 expert annotators (double-blind)
 - Metrics: Faithfulness, Relevance, Informativeness, Conciseness, Coherence (Likert 1–5)

Plan-based Models

- **Plan Generation (PG)**: generates question sequence
- **Summary Generation (SG)**: generates summary answering plan questions

Planning questions

q1: What challenge do large language models face despite their impressive performance on diverse tasks?
q2: What is the aim of this paper regarding large language models?
q3: What is one key finding about LMs' performance with less popular factual knowledge?
q4: How does scaling impact LMs' ability to memorize factual knowledge?
q5: What is the proposed method based on the findings of this paper?

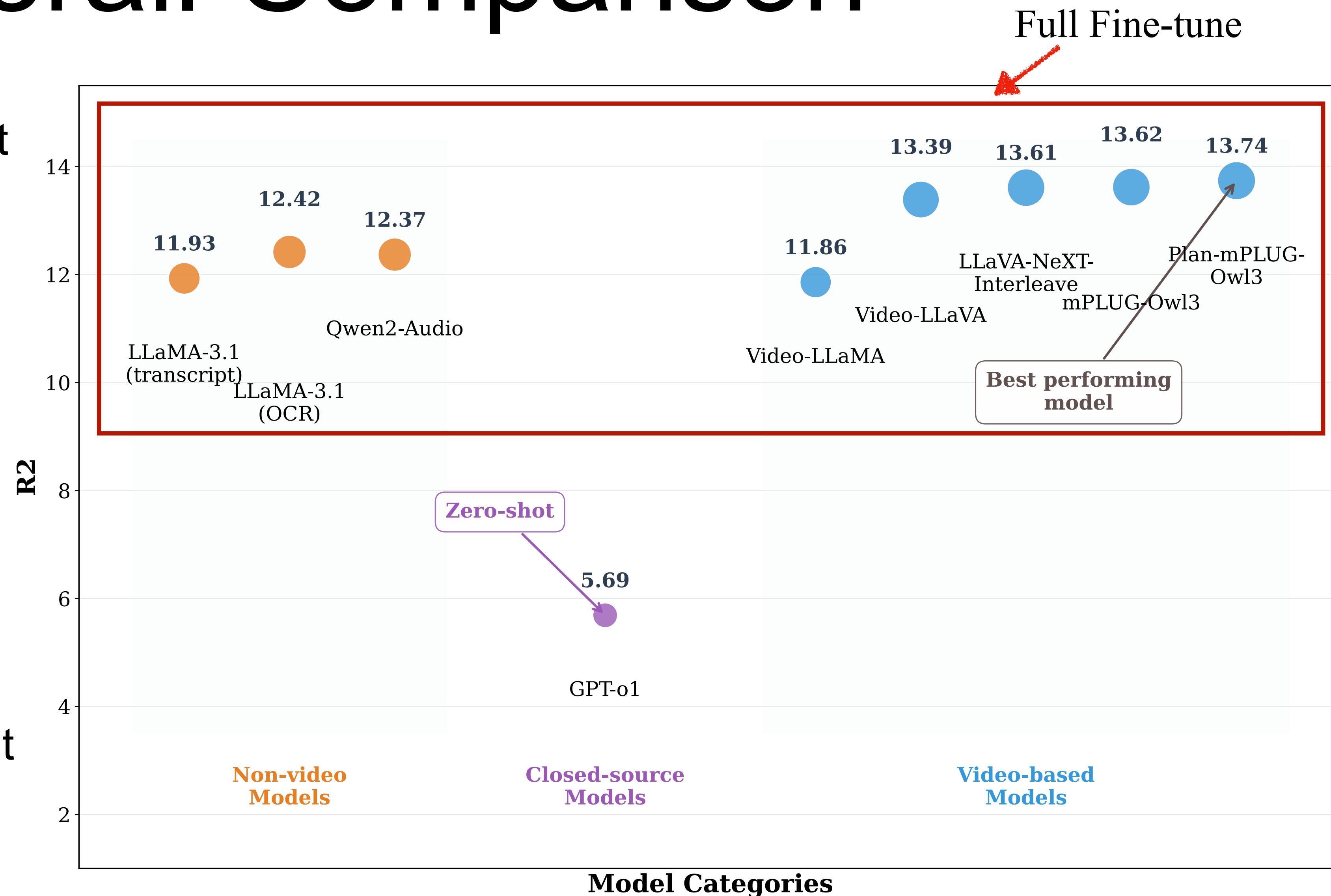
Summary

[Despite their impressive performance on diverse tasks, large language models (LMs) still struggle with tasks requiring rich world knowledge, implying the difficulty of encoding a wealth of world knowledge in their parameters.]^{t1} [This paper aims to understand LMs' strengths and limitations in memorizing factual knowledge, by conducting large-scale knowledge probing experiments on two open-domain entity-centric QA datasets: PopQA, our new dataset with 14k questions about long-tail entities, and EntityQuestions, a widely used open-domain QA dataset.]^{t2} [We find that LMs struggle with less popular factual knowledge, and that retrieval augmentation helps significantly in these cases.]^{t3} [Scaling, on the other hand, mainly improves memorization of popular knowledge, and fails to appreciably improve memorization of factual knowledge in the tail.]^{t4} [Based on those findings, we devise a new method for retrieval-augmentation that improves performance and reduces inference costs by only retrieving non-parametric memories when necessary.]^{t5}

←
Plan Generation
GPT-01

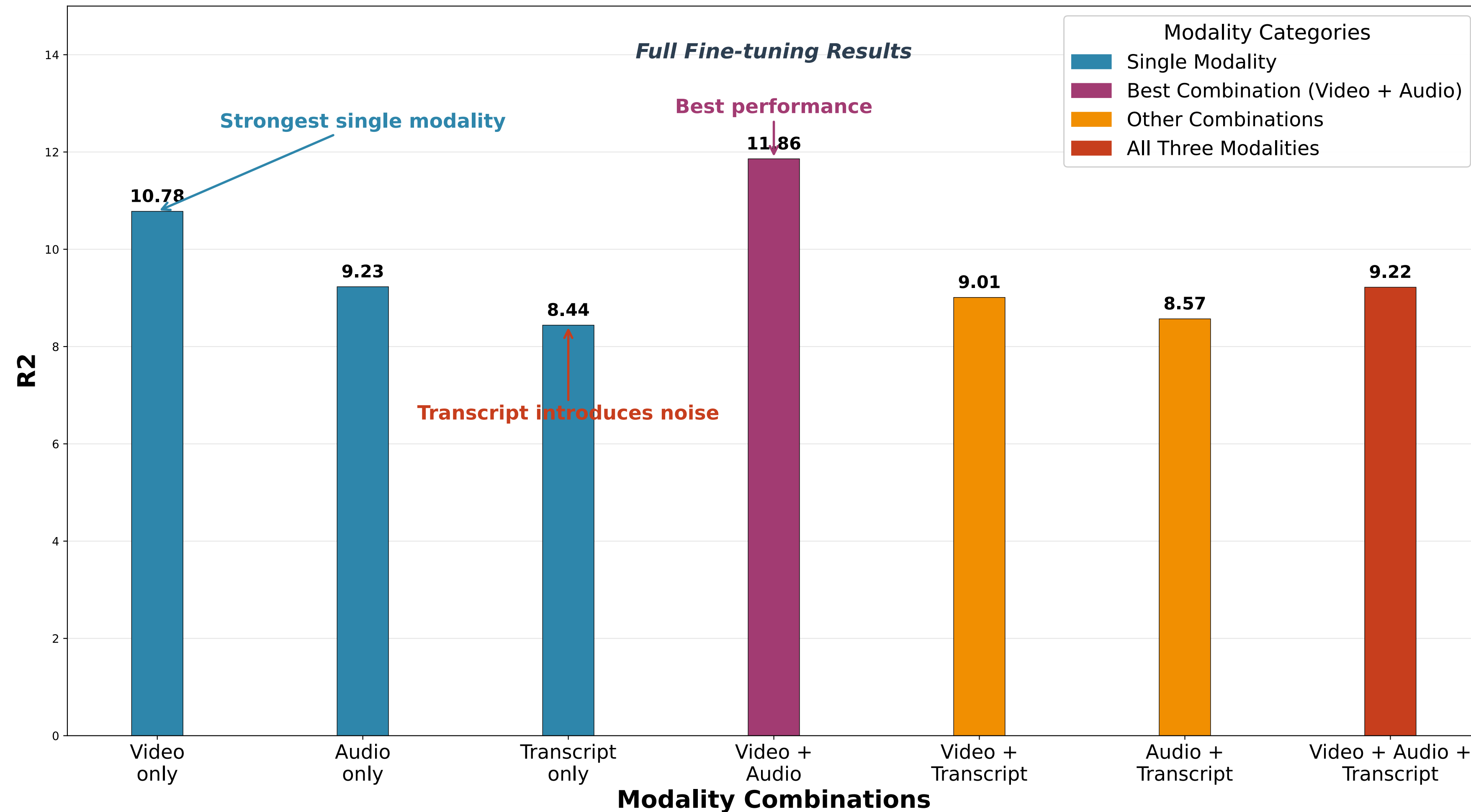
Overall Comparison

- Fine-tuning on in-domain VISTA data yields the largest gains
- Video-based LMMs **outperform** text- and audio-only models
- Closed-source models (e.g., GPT-o1) lead in zero-shot, but open-source models excel after fine-tuning
- Our plan-based method, built on mPLUG-Owl3 achieves **highest overall scores**



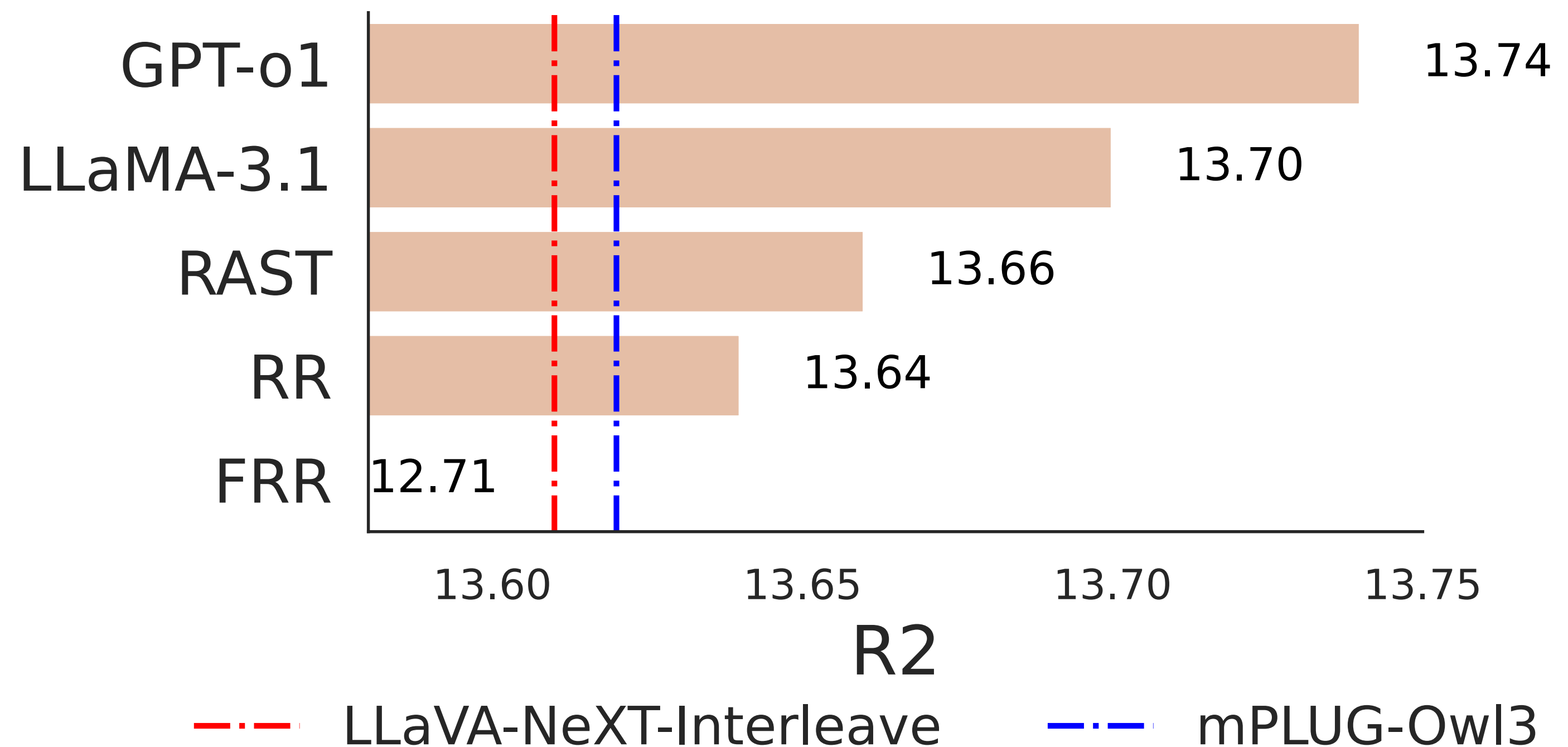
Modalities Matter

- Video is the strongest single modality (spatial-temporal cues)
- Adding audio provides minor gains; transcript (ASR) can introduce noise
- Best results from joint video + audio inputs



Impact of Planning Quality

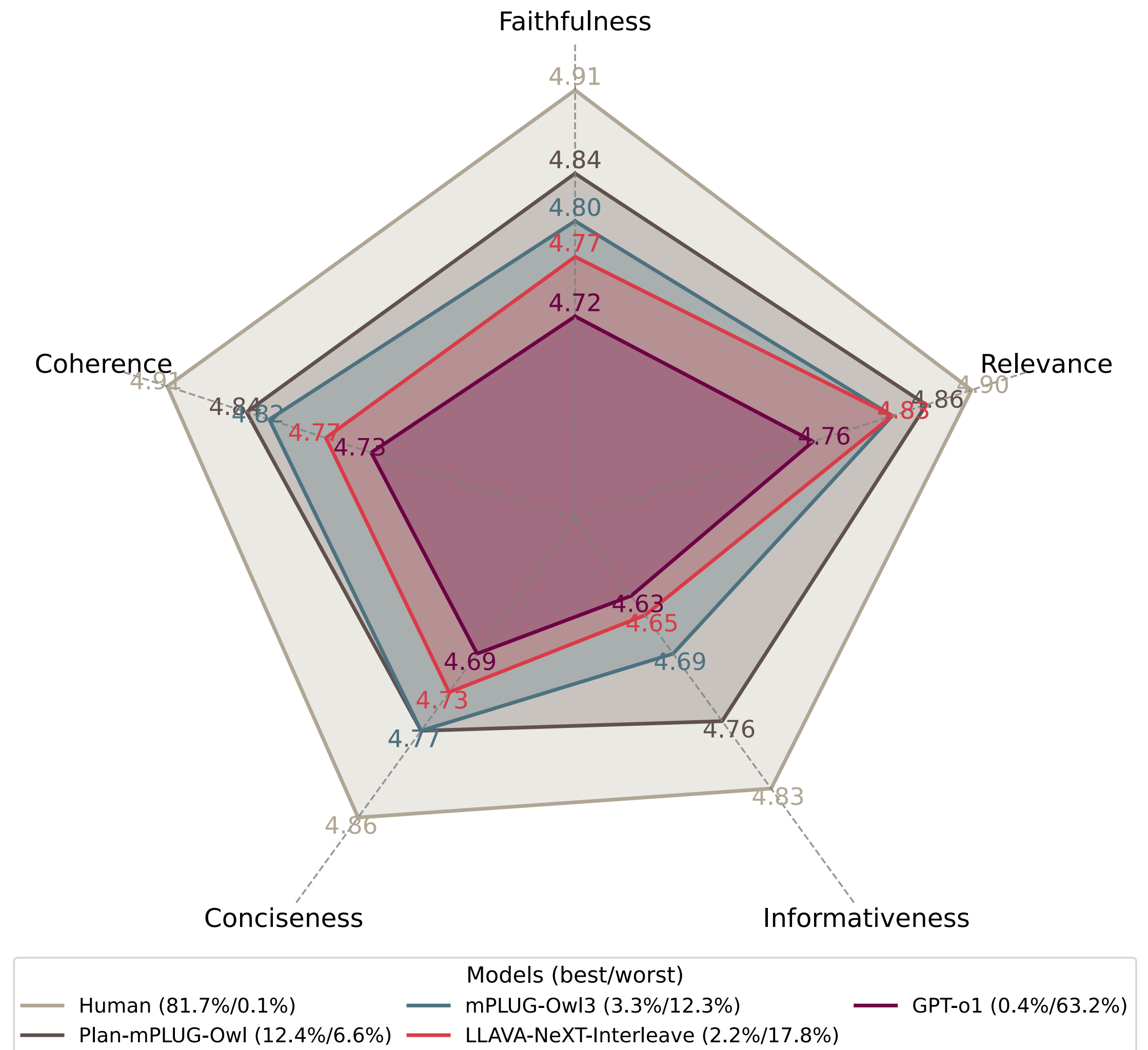
- Higher-quality plans → better summaries
- Noise in planning (irrelevant/random questions) **degrades** summary performance
- Plan-based approach remains robust under moderate noise



RAST (Gou et al., EMNLP 2023) is a SOTA question generation method.
RR = random replacement, FRR = full random replacement

Human Evaluation

- Human-written summaries **outperform** all neural LMMs on faithfulness, informativeness, coherence, etc.
- Plan-based model is **best among neural systems**, but gap with human remains



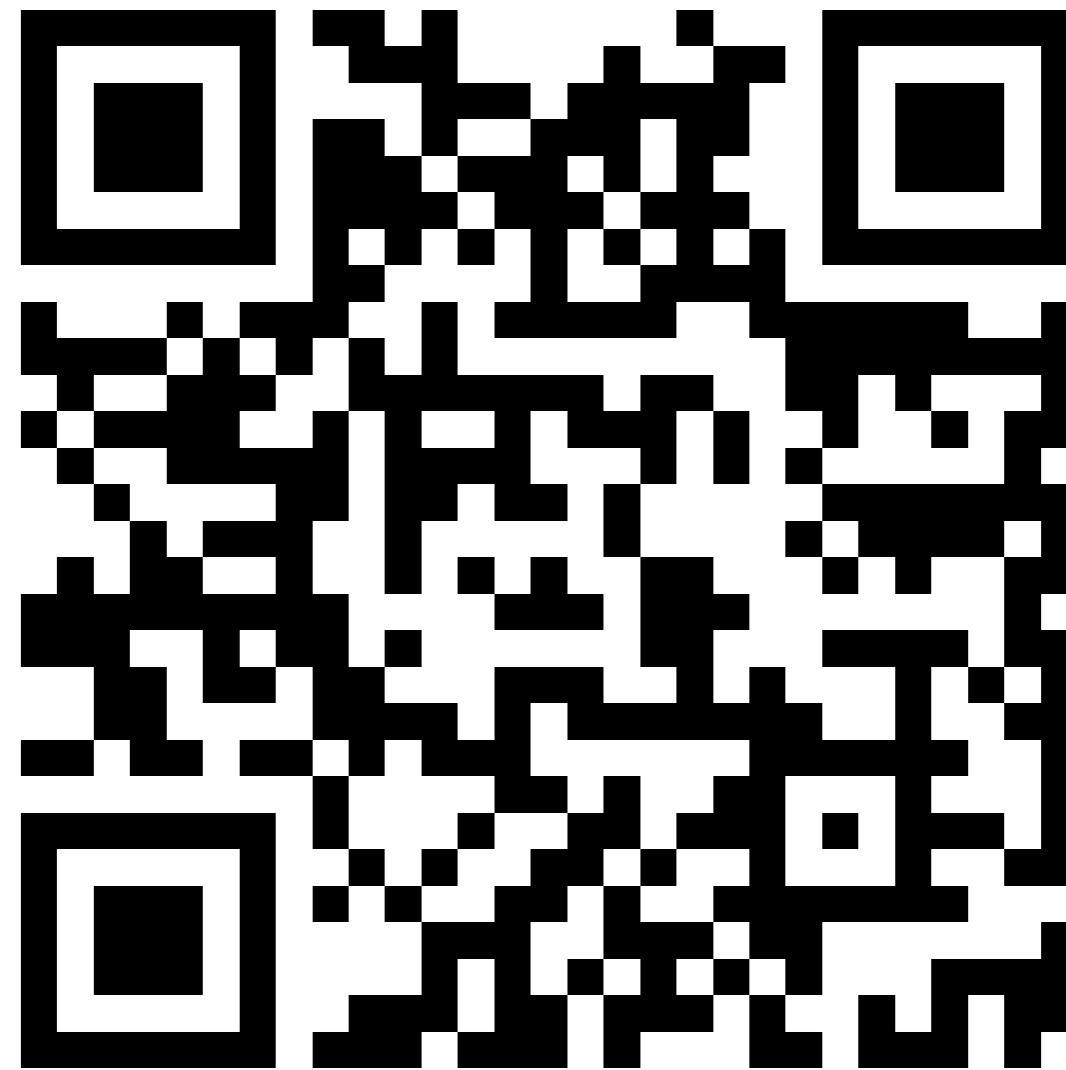
We compare with human performance, the top three finetuned models, and the best-performing closed-source model (under zero-shot setting).

Conclusion

- VISTA establishes a **new benchmark** for scientific video-to-text summarization
- Plan-based models improve **summary quality and factual accuracy**
- **Significant gaps** remain between model and human performance

More Info

- **Data & Code:** <https://dongqi.me/projects/VISTA>
- **Questions:** dongqi.me@gmail.com



Thanks for listening

Q&A



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